

Weekly Progress Report

Weekend of Saturday 18 Sunday 19 July 2026

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Abstract

This report covers the work carried out over Saturday and Sunday. The weekend was spent closing a knowledge gap identified during the Thursday sessions: I mapped out the technical and mathematical background required, built a structured study plan around it, and completed Phase 1 of that plan. Phases 2 and 3 are scheduled for Monday, after which paper review begins.

1. Background: What initiated This Work

After attending the Thursday sessions, it became clear that parts of the discussion sat on foundations I had not yet covered. Rather than continue forward with partial understanding, I decided to stop and address the gap directly.

2. Building the Plan

I began by researching the topic in depth, gathering material from across the internet using both conventional search and AI models to identify what a complete coverage of the subject would actually require, including the equivalent mathematical background that underpins it.

I then fed the collected material to Claude and worked with it to turn a loose pile of sources into a structured, phased learning plan. The result is a plan I can walk through step by step rather than reading at random.

The wider aim is to build the background needed to read papers properly: to follow them at depth, to judge what contribution each one actually makes, and to appreciate the complexity of the problems they tackle. I also expect this foundation, and the mathematics that comes with it, to pay off later when it comes to writing paper.

- **Study plan (Claude conversation):**
<https://claude.ai/share/2c51b57a-d22f-4987-a8cc-5614a9044228>
- **Supporting reference:** Harvard notes covering one of the core topics:
https://people.math.harvard.edu/~knill/teaching/math118/118_dynamicalsystems.pdf

3. Progress to Date

- **Phase 1: Complete.** The foundational material identified in the plan has been worked through.
- **Phases 2 and 3: In progress.** Both are scheduled to be completed on Monday.

4. Plan for the Coming Week

With the foundations in place, the focus shifts from acquiring background knowledge to applying it.

- **Monday:** Finish Phases 2 and 3 of the study plan.
- **Tuesday:** Begin gathering papers and scanning them in depth, producing feedback on each.
- **Tuesday/Wednesday onward:** Assigned tasks proceed in parallel with the paper review.
- **By end of week:** All required background knowledge acquired; 20+ papers scanned, summarised, and reviewed.

This section will be updated in every report sent, so that it always reflects the work actually completed against the plan.